Heart Failure — SCENE 1 — HF CAUSES & PATHOGENESIS

1. MI battering ram (ischemic = #1)

WHAT YOU SEE

A battering ram labeled MI smashing the castle wall

WHAT IT ENCODES

Ischemic heart disease (coronary artery disease / prior MI) is the SINGLE most common cause of HFrEF in the developed world, accounting for roughly two-thirds of cases. Loss of viable myocardium after infarction triggers adverse remodeling: infarct expansion, eccentric dilation, and neurohormonal activation. This is why every new HFrEF patient gets coronary evaluation (angiography or CT) — revascularizing viable/hibernating myocardium can recover function.

BOARD TRAP

WRONG answer: 'idiopathic dilated cardiomyopathy is the most common cause of HFrEF.' Discriminator: idiopathic DCM is a leading NON-ischemic cause, but ischemic heart disease/CAD is overall #1 — always exclude coronary disease before labeling a cardiomyopathy 'idiopathic.'

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2. HFrEF tower (EF ≤40)

WHAT YOU SEE

Tower base placard 'HFrEF EF ≤40'

WHAT IT ENCODES

HFrEF = HF with reduced ejection fraction ≤40%: a problem of systolic contractility with eccentric, dilated (chamber-enlarged) remodeling. This is the EF category with the strongest, most extensive evidence base — all four GDMT pillars (ARNI/ACEi-ARB, beta-blocker, MRA, SGLT2i) carry Class I mortality benefit here, plus device therapy (ICD for EF ≤35%, CRT for EF ≤35% with LBBB/QRS ≥150 ms).

BOARD TRAP

WRONG answer: 'HFrEF means EF in the 41–49% range.' Discriminator: EF 41–49% is HFmrEF (mildly reduced); HFrEF requires EF ≤40% — getting the cutoff wrong changes which GDMT/device indications apply.

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3. HFmrEF (EF 41-49)

WHAT YOU SEE

Central banner 'HFrEF 41-49'

WHAT IT ENCODES

HFmrEF = HF with mildly reduced EF, 41–49%, an intermediate category (formerly 'borderline'). Many of these patients are transitioning — either recovering from HFrEF or declining toward it; trajectory matters. Guidelines give a weaker (Class IIb) recommendation to consider the same GDMT used in HFrEF (especially SGLT2i, which benefit across the full EF spectrum).

BOARD TRAP

WRONG answer: 'HFmrEF patients should not receive any GDMT because the evidence is only for HFrEF.' Discriminator: HFmrEF warrants consideration of HFrEF-type GDMT (notably SGLT2i, with Class I benefit across EF), so withholding all therapy is incorrect.

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4. HFpEF tower (EF ≥50)

WHAT YOU SEE

Right tower base 'HFpEF EF ≥50' with thick concentric walls

WHAT IT ENCODES

HFpEF = HF with preserved EF ≥50%: the heart contracts normally but is STIFF — diastolic dysfunction with concentric hypertrophy and impaired relaxation/filling, so filling pressures rise. Diagnosis combines symptoms + preserved EF + evidence of elevated filling pressures (elevated NT-proBNP, E/e', LA enlargement; H2FPEF or HFA-PEFF scores). SGLT2 inhibitors are the first drug class with proven outcome benefit (EMPEROR-Preserved, DELIVER); treat comorbidities (HTN, AF, obesity).

BOARD TRAP

WRONG answer: 'HFpEF is caused by reduced myocardial contractility.' Discriminator: contractility/EF is PRESERVED in HFpEF — the defect is diastolic (stiff ventricle, impaired relaxation/filling), not systolic pump failure.

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5. HTN cannons → HFpEF

WHAT YOU SEE

Cannons labeled HTN firing pressure at the HFpEF tower

WHAT IT ENCODES

Hypertension is the leading cause/risk factor for HFpEF and the second most common cause of HF overall: chronic pressure overload → concentric LV hypertrophy → a stiff, poorly relaxing ventricle with elevated diastolic filling pressures. Aggressive BP control (target <130/80) is the key preventive/therapeutic lever (SPRINT showed intensive BP lowering reduced incident HF).

BOARD TRAP

WRONG answer: 'hypertension is the #1 cause of heart failure overall.' Discriminator: HTN is a MAJOR contributor (and the top HFpEF driver), but ischemic heart disease/CAD is the single most common cause of HF overall — HTN ranks second.

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6. Pressure overload → concentric

WHAT YOU SEE

Worker building thick 'CONCENTRIC' wall under PRESSURE sign

WHAT IT ENCODES

PRESSURE overload (hypertension, aortic stenosis) forces the ventricle to generate higher systolic pressure → sarcomeres added in PARALLEL → CONCENTRIC hypertrophy: walls thicken, cavity stays normal/small (increased wall-thickness-to-radius ratio). This normalizes wall stress (Laplace) initially but produces a stiff chamber → diastolic dysfunction/HFpEF. Mnemonic: Pressure = Parallel sarcomeres = thick walls.

BOARD TRAP

WRONG answer: 'pressure overload (e.g., aortic stenosis) produces eccentric chamber dilation.' Discriminator: pressure overload → concentric (parallel sarcomeres, thick walls); it is VOLUME overload that produces eccentric dilation — do not swap them.

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7. Volume overload → eccentric

WHAT YOU SEE

Workers labeled VOLUME building 'ECCENTRIC' dilated wall

WHAT IT ENCODES

VOLUME overload (mitral regurgitation, aortic regurgitation, large left-to-right shunts) increases diastolic wall stress → sarcomeres added in SERIES → ECCENTRIC hypertrophy: the chamber DILATES with proportional wall thinning. Over time this progresses to systolic dysfunction/HFrEF. Mnemonic: Volume = series sarcomeres = dilated chamber.

BOARD TRAP

WRONG answer: 'chronic mitral or aortic regurgitation causes concentric hypertrophy with a small cavity.' Discriminator: regurgitant (volume) lesions cause eccentric dilation (series sarcomeres, enlarged cavity); concentric hypertrophy is the response to pressure overload.

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8. Neurohormonal: SNS / RAAS

WHAT YOU SEE

Red muscleman 'SNS', monk holding RENIN / ANG II vials, fetal genes

WHAT IT ENCODES

Falling cardiac output triggers maladaptive neurohormonal activation: the sympathetic nervous system (↑catecholamines → tachycardia, ↑afterload, myocyte toxicity/apoptosis, arrhythmia) and the RAAS (angiotensin II → vasoconstriction; aldosterone → Na/water retention and myocardial FIBROSIS). Reactivation of the FETAL gene program drives pathologic remodeling. This cascade is the rationale for GDMT — beta-blockers, ACEi/ARB/ARNI, and MRAs each BLOCK an arm and improve survival.

BOARD TRAP

WRONG answer: 'inotropes that boost sympathetic drive improve long-term survival in chronic HFrEF.' Discriminator: chronic neurohormonal activation is MALADAPTIVE — drugs that ANTAGONIZE it (beta-blockers, RAAS inhibitors) prolong life, whereas sympathomimetic inotropes increase mortality in chronic HF.

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9. Natriuretic peptides / neprilysin

WHAT YOU SEE

Doves (ANP/BNP) and a spider labeled NEPRILYSIN at the top

WHAT IT ENCODES

Natriuretic peptides (ANP from atria, BNP from ventricles) are the heart's protective counter-regulators — released in response to wall stretch, they cause vasodilation, natriuresis/diuresis, and oppose RAAS. Neprilysin is the enzyme that DEGRADES them. ARNI (sacubitril/valsartan) pairs a neprilysin inhibitor (boosts protective peptides) with an ARB (blocks Ang II); PARADIGM-HF showed ARNI beat enalapril for CV death/HF hospitalization.

BOARD TRAP

WRONG answer: 'combine sacubitril/valsartan (ARNI) with an ACE inhibitor for additive benefit.' Discriminator: ARNI + ACEi causes dangerous ANGIOEDEMA — require a ≥36-hour washout when switching from an ACEi to ARNI; the two are never given together.

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10. LBBB → CRT

WHAT YOU SEE

Banner 'LBBB' with 'CRT' flag on the castle

WHAT IT ENCODES

A wide LBBB produces electrical and mechanical DYSSYNCHRONY (the septum and lateral wall contract out of phase), worsening efficiency in HFrEF. Cardiac resynchronization therapy (CRT, biventricular pacing) re-coordinates contraction. Best candidates (Class I): EF ≤35%, LBBB morphology, QRS ≥150 ms, NYHA II–IV on GDMT — these gain the most mortality/symptom benefit.

BOARD TRAP

WRONG answer: 'CRT benefits all HFrEF patients with a wide QRS regardless of morphology.' Discriminator: CRT benefit is greatest with true LBBB and QRS ≥150 ms; non-LBBB or narrow QRS (<130 ms) patients derive little/no benefit (and CRT may harm in QRS <130 ms — EchoCRT).

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11. High-output HF causes

WHAT YOU SEE

Framed 'HIGH OUTPUT' picture: obesity, anemia, beriberi (thiamine), cirrhosis, AV fistula, hyperthyroid

WHAT IT ENCODES

High-output HF = the heart pumps a HIGH cardiac output yet still fails to meet metabolic demand or compensate for reduced systemic vascular resistance/shunting. Causes (mnemonic AABBCC-PT): Anemia (severe/chronic), AV fistula (incl. dialysis access), Beriberi (thiamine/B1 deficiency — 'wet beriberi'), Cirrhosis, Carcinoid, Paget disease of bone, Thyrotoxicosis, plus obesity/pregnancy. Look for warm extremities, bounding pulses, wide pulse pressure.

BOARD TRAP

WRONG answer: 'high-output heart failure means the cardiac output is low.' Discriminator: in high-output HF the cardiac output is HIGH (often >8 L/min) — failure occurs because demand/shunting outstrips even that elevated output; warm extremities and bounding pulses distinguish it from low-output (cold) HF.

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12. Infiltrative / toxic cardiomyopathy

WHAT YOU SEE

SARCOID/granuloma cross, ALDO/MRA bottle, CHEMO/OH (alcohol) bottle at base

WHAT IT ENCODES

Non-ischemic cardiomyopathies to know: INFILTRATIVE/restrictive — amyloidosis (low-voltage ECG with paradoxical LVH wall thickening; transthyretin-type treated with tafamidis), sarcoidosis (granulomas → heart block/VT, treat with steroids ± device), hemochromatosis. TOXIC/dilated — anthracycline chemotherapy (dose-dependent, e.g., doxorubicin) and trastuzumab (reversible); ALCOHOL (chronic, dose-dependent, may reverse with abstinence). Takotsubo (stress) cardiomyopathy: apical ballooning after emotional/physical stress, usually reversible.

BOARD TRAP

WRONG answer: 'discordance between low ECG voltage and a thick-walled ventricle on echo is just measurement error.' Discriminator: low voltage WITH increased wall thickness is the classic fingerprint of cardiac AMYLOIDOSIS (infiltration, not true hypertrophy) — a key board pattern, not an artifact.

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13. Infectious causes (Chagas, HIV)

WHAT YOU SEE

Skeleton knights labeled CHAGAS, HIV, SLE, myocarditis on the left

WHAT IT ENCODES

Infective/inflammatory dilated cardiomyopathy: Chagas disease (Trypanosoma cruzi — leading infectious DCM cause in Latin America; classically apical aneurysm, RBBB + left anterior fascicular block, megaesophagus/megacolon), HIV cardiomyopathy, VIRAL myocarditis (Coxsackie B, parvovirus B19, HHV-6; can follow a viral prodrome and present as new HF, arrhythmia, or sudden death), and autoimmune (SLE). PERIPARTUM cardiomyopathy: HFrEF in the last month of pregnancy through ~5 months postpartum, often recovers but may recur with subsequent pregnancies.

BOARD TRAP

WRONG answer: 'a young patient with a recent viral illness and new HF/arrhythmia most likely has a chronic ischemic cardiomyopathy.' Discriminator: new HF in a young person after a viral prodrome points to acute VIRAL MYOCARDITIS (consider endomyocardial biopsy/MRI), not CAD — and giant-cell myocarditis is a fulminant variant needing urgent recognition.